Presented by John Speaks

STRATEGIC DEFAULT PREVENTION



AGENDA

- Introduction
- What is credit risk, and why is it a problem for us?
- Methodology
- Plan of Action
- Understanding the Data
- Solutioning through Modeling
- Results and Implications
- Recommendations and Next Steps

WELCOME

Hello, I'm John Speaks, a Master's Data Analytic Graduate from SNHU.

This presentation is meant to support my Capstone Analytical Project within the program.



RISKY BUSINESS

CREDIT RISK

- What is it?
  - The probability that a borrower will fail to settle their financial obligations (Paz et al., 2025)
- Impact on the business
  - Borrowers failing to repay or settle their debts can be a source of financial loss for GE
  - The inability to accurately identify borrowers who present a greater credit risk is a major business concern

How can the employer accurately predict high-risk borrowers and minimize losses from credit default?



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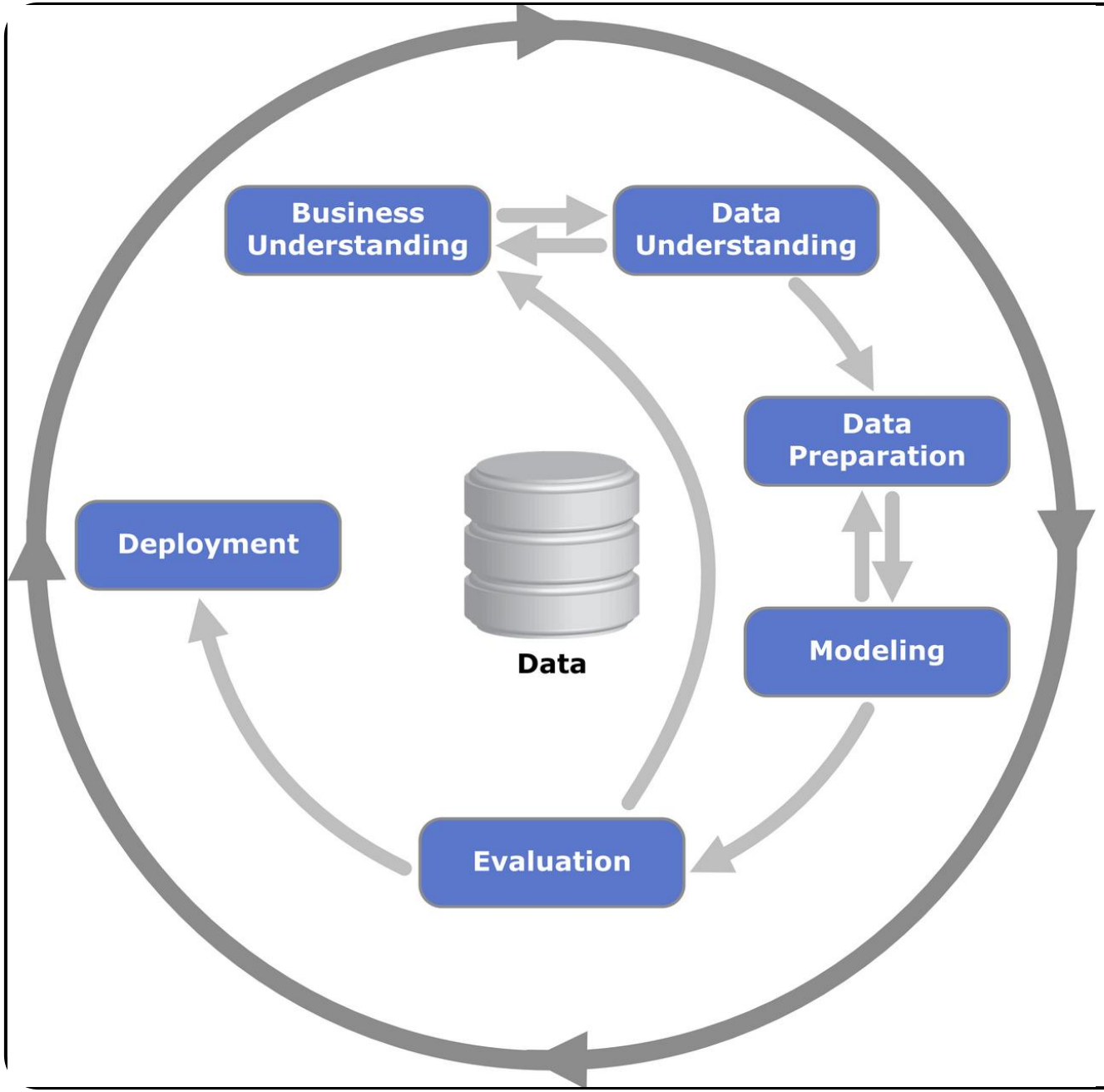
Project Objective: Create a Model that can predict default with an accuracy of

70% or greater

WHERE DO WE GO FROM HERE?

CROSS-INDUSTRY STANDARD PROCESS FOR DATA MINING (CRISP – DM)

PHASE	TITLE	DESCRIPTION
1	Business Understanding	• Focused on understanding of project goals from a business perspective • Translating the requirements into an analytical model
2	Data Understanding	• Gathering the data and becoming more familiar with it
3	Data Preparation	• Making necessary additions, deletions, and transformations in order to prepare the data for modeling
4	Modeling	• Conducting Analysis using chosen model
5	Evaluation	• Reviewing and assessing model performance in relation to business objectives
6	Deployment	• Moving model into production to support decision-making



CRISP – DM DIAGRAM

Please note that the process is bi-directional at multiple stages

ANALYTICAL PLAN

CRISP-DM ANALYTICAL PLAN

- **Business Understanding:** GE relies on credit risk assessment to prevent financial losses from lending to borrowers that lack creditworthiness. To mitigate this risk, the analyst seeks to build a model to predict the likelihood of borrower default. By accurately predicting high-risk borrowers, GE can make confident lending decisions.
- **Data Understanding:** For this portion of the project, I will review the data to understand the different data types present within the dataset (categorical vs. numerical). Also, this stage involves using descriptive statistics (min, mean, max, etc.) and visualizations, such as histograms, to summarize variables and assess the data's shape to identify patterns.
- **Data Preparation:** During this phase, I will utilize R to identify duplicate records, null values, outliers, and potential erroneous entries. Once data inaccuracies are properly remedied, I will use principal component analysis (PCA) to reduce data dimensions (variables) prior to analysis (B. Wu et al., 2025). This helps reduce unnecessary noise that may negatively impact the model.
- **Modeling:** I have chosen Logistic Regression for its strength as a classifier. If tuned properly, it will help us take an analytical approach to determine the likelihood of borrower default based on the most crucial dimensions of each record.
- **Evaluation:** During this phase, tools such as a confusion matrix will provide insight into the effectiveness of predicting. Key metrics will include accuracy, recall, precision, and F1-Score. An ROC-AUC Curve will also be useful in effective model tuning. By comparing these values, I will determine whether the model reaches acceptable overall accuracy thresholds (70% or greater) **Deployment:** During this phase, the model would be used within GE's decision-making process. If a borrower is flagged as high risk of default, steps should be taken to mitigate that risk. Recommendations may include limiting the amount of credit awarded, denying credit, or charging higher interest rates to reflect the acceptance of default risk.

UNDERSTANDING THE DATA

Review of the Available Data and Preparation Steps

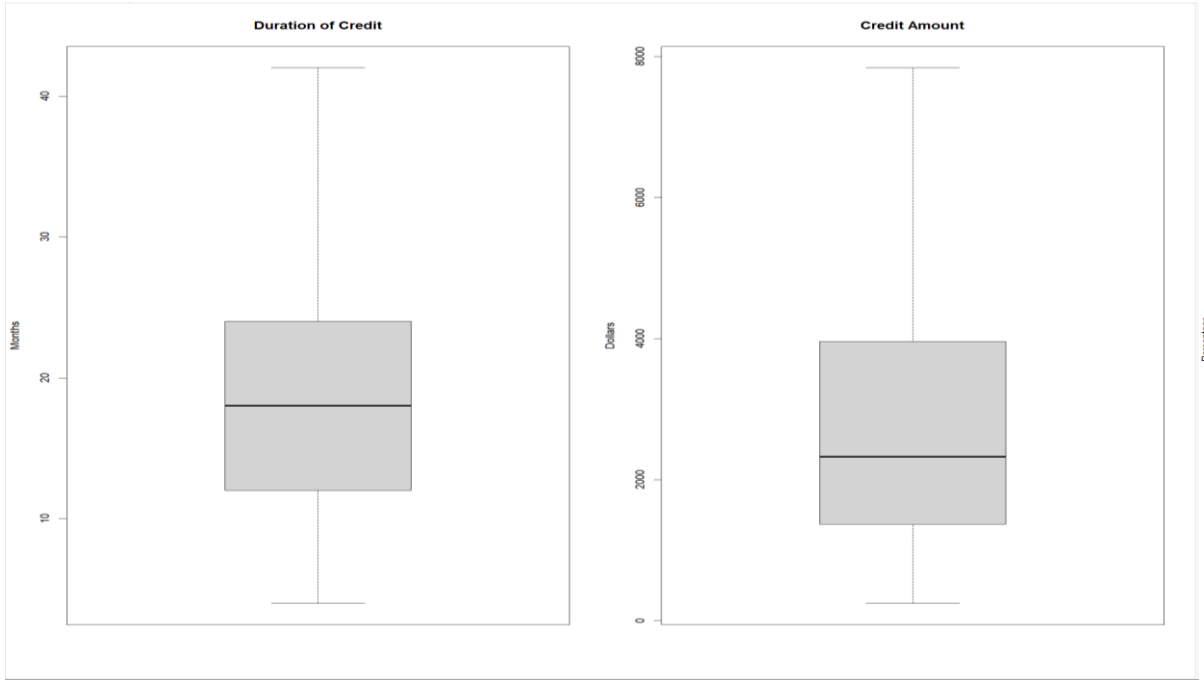
DATA UNDERSTANDING

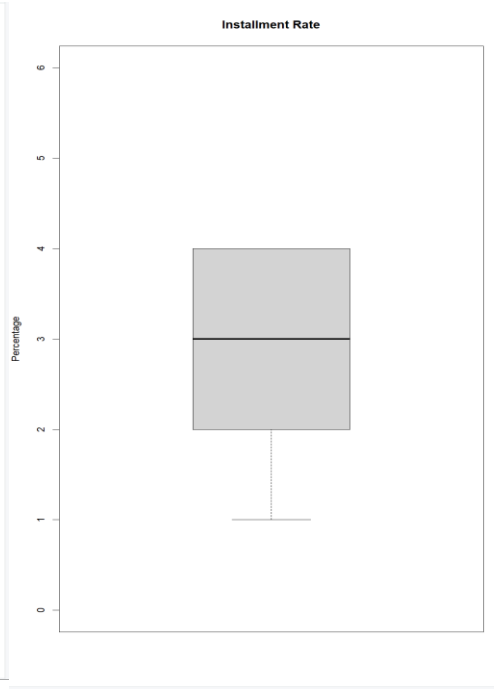
- Data was collected by GE over the course of doing business
- Sample includes 800 observations
- 32 Features per observation
  - 30 Independent variables
  - 1 Target Variable
  - 1 Unique Identifier (OBS)
- Data is relatively clean
  - 0 Duplicate Records
  - 0 Null Values
  - Some outliers
- Numerical, Categorical, and Binary Data Types
  - 6 Numerical
  - 20 Binary
  - 5 Categorical

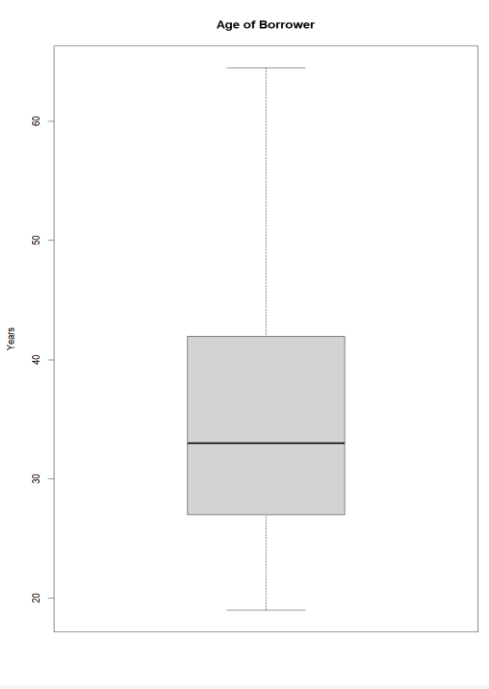
```
> str(gecred)
'data.frame':   800 obs. of  32 variables:
 $ OBS.             : int  1 2 3 4 5 6 7 8 9 10 ...
 $ CHK_ACCT         : int  0 1 3 0 0 3 3 1 3 1 ...
 $ DURATION         : int  6 48 12 42 24 36 24 36 12 30 ...
 $ HISTORY          : int  4 2 4 2 3 2 2 2 2 4 ...
 $ NEW_CAR          : int  0 0 0 0 1 0 0 0 0 1 ...
 $ USED_CAR         : int  0 0 0 0 0 0 0 1 0 0 ...
 $ FURNITURE        : int  0 0 0 1 0 0 1 0 0 0 ...
 $ RADIO.TV         : int  1 1 0 0 0 0 0 0 1 0 ...
 $ EDUCATION        : int  0 0 1 0 0 1 0 0 0 0 ...
 $ RETRAINING       : int  0 0 0 0 0 0 0 0 0 0 ...
 $ AMOUNT           : int  1169 5951 2096 7882 4870 9055 2835 6948 3059 5234 ...
 $ SAV_ACCT         : int  4 0 0 0 0 4 2 0 3 0 ...
 $ EMPLOYMENT       : int  4 2 3 3 2 2 4 2 3 0 ...
 $ INSTALL_RATE     : int  4 2 2 2 3 2 3 2 2 4 ...
 $ MALE_DIV         : int  0 0 0 0 0 0 0 0 1 0 ...
 $ MALE_SINGLE      : int  1 0 1 1 1 1 1 1 0 0 ...
 $ MALE_MAR_or_WID  : int  0 0 0 0 0 0 0 0 0 1 ...
 $ CO.APPLICANT     : int  0 0 0 0 0 0 0 0 0 0 ...
 $ GUARANTOR        : int  0 0 0 1 0 0 0 0 0 0 ...
 $ PRESENT_RESIDENT: int  4 2 3 4 4 4 4 2 4 2 ...
 $ REAL_ESTATE      : int  1 1 1 0 0 0 0 0 1 0 ...
 $ PROP_UNKN_NONE   : int  0 0 0 0 1 1 0 0 0 0 ...
 $ AGE              : int  67 22 49 45 53 35 53 35 61 28 ...
 $ OTHER_INSTALL    : int  0 0 0 0 0 0 0 0 0 0 ...
 $ RENT             : int  0 0 0 0 0 0 0 0 1 0 ...
 $ OWN_RES          : int  1 1 1 0 0 0 1 0 1 1 ...
 $ NUM_CREDITS      : int  2 1 1 1 2 1 1 1 1 2 ...
 $ JOB              : int  2 2 1 2 2 1 2 3 1 3 ...
 $ NUM_DEPENDENTS   : int  1 1 2 2 2 2 1 1 1 1 ...
 $ TELEPHONE        : int  1 0 0 0 0 1 0 1 0 0 ...
 $ FOREIGN          : int  0 0 0 0 0 0 0 0 0 0 ...
 $ DEFAULT          : int  0 1 0 0 1 0 0 0 0 1 ...
```

DATA UNDERSTANDING

```
> summary(gecred)
      OBS.          CHK_ACCT         DURATION         HISTORY          NEW_CAR          USED_CAR         FURNITURE          RADIO.TV         EDUCATION        RETRAINING         AMOUNT          SAV_ACCT       EMPLOYMENT     INSTALL_RATE
 Min.   :   1.0   Min.   :0.000   Min.   : 4.00   Min.   :0.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.00000   Min.   :0.00000   Min.   :  250   Min.   :0.000   Min.   :0.000   Min.   :1.000
 1st Qu.: 400.8   1st Qu.:0.000   1st Qu.:12.00   1st Qu.:2.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.: 1374   1st Qu.:0.000   1st Qu.:2.000   1st Qu.:2.000
 Median : 600.5   Median :1.000   Median :18.00   Median :2.000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.00000   Median :0.00000   Median : 2326   Median :0.000   Median :2.000   Median :3.000
 Mean   : 576.0   Mean   :1.567   Mean   :21.01   Mean   :2.539   Mean   :0.2437   Mean   :0.1037   Mean   :0.1787   Mean   :0.2725   Mean   :0.05375   Mean   :0.09375   Mean   : 3272   Mean   :1.109   Mean   :2.356   Mean   :2.965
 3rd Qu.: 800.2   3rd Qu.:3.000   3rd Qu.:24.00   3rd Qu.:4.000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.: 3960   3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:4.000
 Max.   :1000.0   Max.   :3.000   Max.   :72.00   Max.   :4.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :1.00000   Max.   :1.00000   Max.   :18424   Max.   :4.000   Max.   :4.000   Max.   :4.000
    MALE_DIV       MALE_SINGLE     MALE_MAR_or_WID     CO.APPLICANT      GUARANTOR       PRESENT_RESIDENT  REAL_ESTATE      PROP_UNKN_NONE       AGE         OTHER_INSTALL        RENT            OWN_RES         NUM_CREDITS         JOB
 Min.   :0.00000   Min.   :0.0000   Min.   :0.00000   Min.   :0.0000   Min.   :0.0000   Min.   :1.000   Min.   :0.0000   Min.   :0.000   Min.   :19.00   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :1.000   Min.   :0.000
 1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:27.00   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.000   1st Qu.:2.000
 Median :0.00000   Median :1.0000   Median :0.00000   Median :0.0000   Median :0.0000   Median :3.000   Median :0.0000   Median :0.000   Median :33.00   Median :0.0000   Median :0.0000   Median :0.0000   Median :1.000   Median :2.000
 Mean   :0.04375   Mean   :0.5475   Mean   :0.09375   Mean   :0.0425   Mean   :0.0525   Mean   :2.868   Mean   :0.2725   Mean   :0.155   Mean   :35.56   Mean   :0.1862   Mean   :0.1837   Mean   :0.7063   Mean   :1.399   Mean   :1.893
 3rd Qu.:0.00000   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:4.000   3rd Qu.:1.0000   3rd Qu.:0.000   3rd Qu.:42.00   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:2.000   3rd Qu.:2.000
 Max.   :1.00000   Max.   :1.0000   Max.   :1.00000   Max.   :1.0000   Max.   :1.0000   Max.   :4.000   Max.   :1.0000   Max.   :1.000   Max.   :75.00   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000   Max.   :4.000   Max.   :3.000
 NUM_DEPENDENTS     TELEPHONE        FOREIGN         DEFAULT
 Min.   :1.000   Min.   :0.000   Min.   :0.00   Min.   :0.0000
 1st Qu.:1.000   1st Qu.:0.000   1st Qu.:0.00   1st Qu.:0.0000
 Median :1.000   Median :0.000   Median :0.00   Median :0.0000
 Mean   :1.155   Mean   :0.395   Mean   :0.03   Mean   :0.3063
 3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:0.00   3rd Qu.:1.0000
 Max.   :2.000   Max.   :1.000   Max.   :1.00   Max.   :1.0000
```



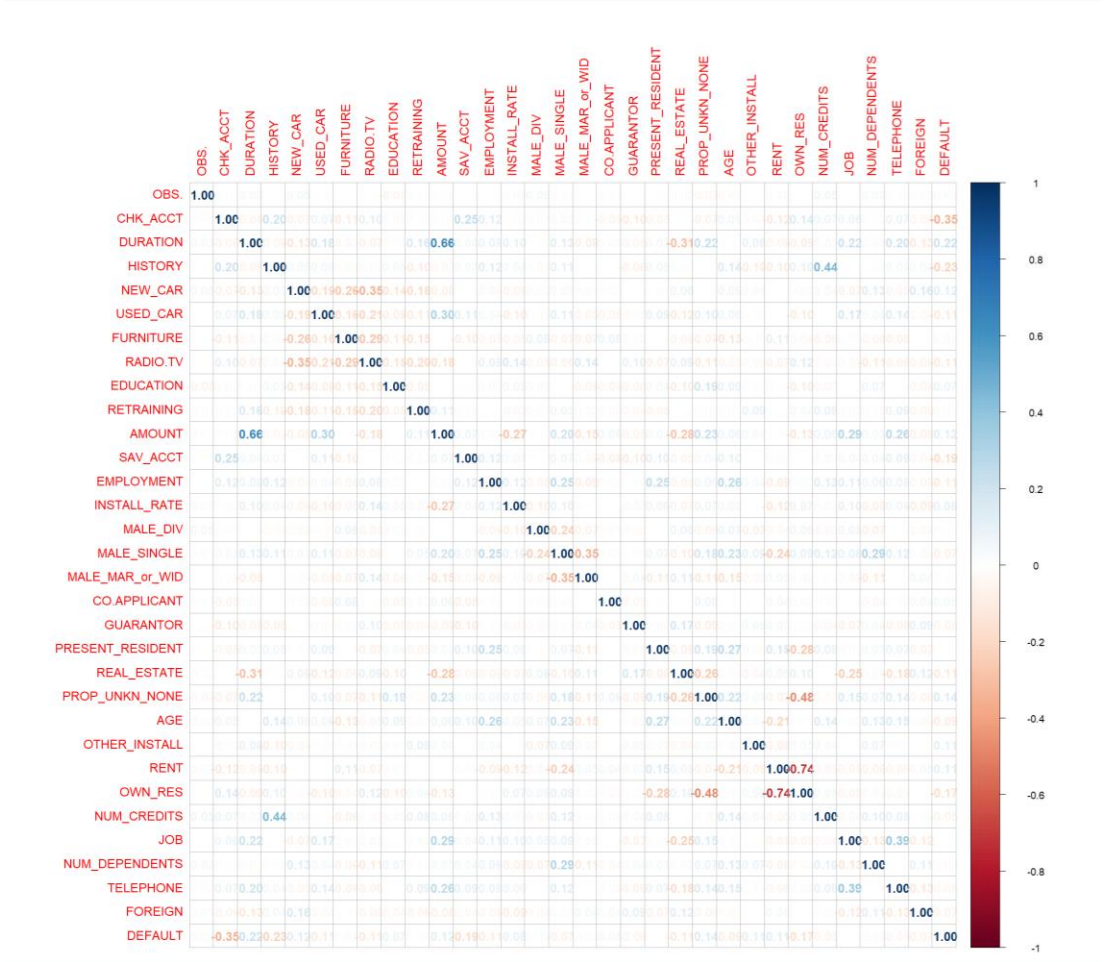




CORRELATION MATRIX

Highly Correlated Variables:

- Own_res & Rent: -0.74
- Amount & Duration: 0.66
- Num_Credit & History: 0.44
- Prop_Unkn_None & Own_Res: -0.48

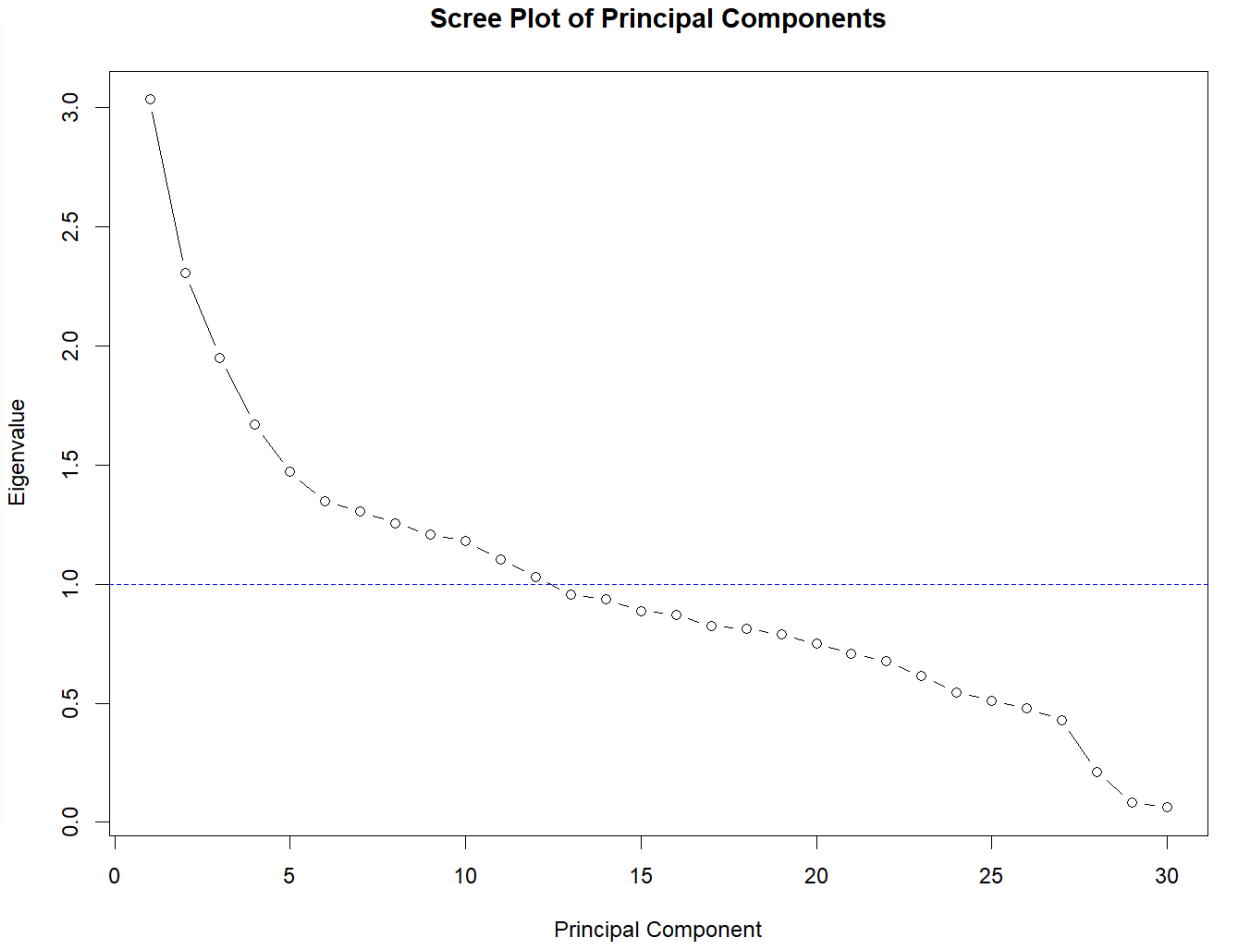


DATA PREPARATION

- Removal of Outliers
  - Prevention of extreme values having undue influence within the model
- Min-Max Scaling
  - Utilized to align dimensions with different scales, and prevent values with larger ranges (i.e., Education vs. Amount) from over-influencing the classification results within a model (Saichon, 2022)
  - Places all values within a range of 0 – 1
- Principal Component Analysis (PCA)
  - Helps to reduce dimensionality while maintaining important statistical information necessary for model training
  - Solves the issue of collinearity by creating 30 new uncorrelated components
  - Scree plot is used to determine the optimal number of components using the Maximum Eigenvalue Ratio Criterion (MERC)
    - MERC calls for retaining all components with Eigenvalues greater than 1 (Luo et al., 2009)

PCA SUMMARY AND SCREE PLOT

```
> summary(pc.cred)
Importance of components:
                          PC1     PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     1.7422 1.51858 1.39664 1.29180 1.21272 1.1606 1.14216
Proportion of Variance 0.1012 0.07687 0.06502 0.05563 0.04902 0.0449 0.04348
Cumulative Proportion  0.1012 0.17804 0.24306 0.29869 0.34771 0.3926 0.43610
                          PC8     PC9    PC10    PC11    PC12    PC13    PC14
Standard deviation     1.12013 1.09916 1.08706 1.05049 1.01523 0.97702 0.96719
Proportion of Variance 0.04182 0.04027 0.03939 0.03678 0.03436 0.03182 0.03118
Cumulative Proportion  0.47792 0.51819 0.55758 0.59437 0.62872 0.66054 0.69172
                         PC15    PC16    PC17    PC18    PC19    PC20    PC21
Standard deviation     0.94221 0.93315 0.90757 0.90148 0.88756 0.86578 0.84069
Proportion of Variance 0.02959 0.02903 0.02746 0.02709 0.02626 0.02499 0.02356
Cumulative Proportion  0.72132 0.75034 0.77780 0.80489 0.83114 0.85613 0.87969
                         PC22    PC23    PC24    PC25    PC26    PC27    PC28
Standard deviation     0.82170 0.78282 0.73904 0.71493 0.69105 0.65467 0.45989
Proportion of Variance 0.02251 0.02043 0.01821 0.01704 0.01592 0.01429 0.00705
Cumulative Proportion  0.90220 0.92262 0.94083 0.95787 0.97378 0.98807 0.99512
                         PC29    PC30
Standard deviation     0.28780 0.25206
Proportion of Variance 0.00276 0.00212
Cumulative Proportion  0.99788 1.00000
```



THE SOLUTION

Our Model

LOGISTIC REGRESSION (LR)

Why LR?

- Logistic Regression is a well-established solution. Known for its strength in representing relationships between the dependent binary variable (DEFAULT) and other independent features (Selim et al., 2024)
- The LR model will estimate the probability of default given the values present within the data features (principal components), and this will enable the model to make accurate predictions after training

How will we evaluate model performance?

- We will measure the model's ability to identify those prone to default with an acceptable level of accuracy (70% or greater)

Key Metrics:

- **Precision:** proportion of correct positive predictions in relation to the total number of positive predictions.
- **Recall:** Proportion of correct positive predictions outputted by the model.
- **F1-Score:** The harmonic mean of the Precision and Recall metrics. This will help ensure balance across our previous metrics, enabling better evaluation of model results.
- **Accuracy:** Proportion of correct default predictions across both positive and negative instances. This will help determine the overall effectiveness of the model.
- **Receiver-operating characteristic Curve (ROC):** Depicts the true positive rate (TPR) and Specificity (1 - False Positive Rate (FPR)) at various thresholds. A perfect model will have TPR of 1 and FPR of 0.
    - **Area Under the Curve (AUC):** Describes whether the model will correctly rank a positive example higher than a negative example given both (selected randomly)

KEY ASSUMPTIONS OF LR

- **Linearity**: Relationship between the independent and dependent variables must be linear
  - Hosmer-Lemeshow tests suggest the model is a good fit for data (p-value = .8974)
- **Independence**: The observations must be independent of each other
  - 800 observations taken independently of each other
- **Multicollinearity**: Independent variables must not be highly correlated
  - Principal Component Analysis creates new uncorrelated components
- **Lack of Outliers:** The data must be free of outliers that could strongly influence the model
  - Outliers were removed in the data preparations phase

Source: Stoltzfus (2011)

```
        Hosmer and Lemeshow goodness of fit (GOF) test

data:  pca.cred3$DEFAULT, fitted(logit.cred3)
X-squared = 3.5233, df = 8, p-value = 0.8974
```

MODEL RESULTS

Evaluation of the outputs from key metrics

INITIAL MODEL WITH 12 PRINCIPAL COMPONENTS

- PC11 & PC12 do not seem to be significant indicators. We will remove them to reduce the dimensionality of the data even further.

```
Call:
glm(formula = DEFAULT ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6 + PC7 +
    PC8 + PC9 + PC10 + PC11 + PC12, family = "binomial", data = pca.cred)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.11600    0.09883 -11.292  < 2e-16 ***
PC1         -0.04464    0.05112  -0.873  0.38260
PC2         -0.55457    0.06507  -8.522  < 2e-16 ***
PC3          0.00809    0.06238   0.130  0.89682
PC4         -0.28877    0.07038  -4.103 4.08e-05 ***
PC5          0.55544    0.08012   6.933 4.13e-12 ***
PC6         -0.10335    0.07437  -1.390  0.16462
PC7         -0.06651    0.07769  -0.856  0.39196
PC8          0.52370    0.08352   6.270 3.61e-10 ***
PC9         -0.32293    0.08001  -4.036 5.43e-05 ***
PC10        -0.27227    0.08335  -3.267  0.00109 **
PC11        -0.06227    0.08223  -0.757  0.44891
PC12        -0.02757    0.08501  -0.324  0.74568
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 985.71  on 799  degrees of freedom
Residual deviance: 775.35  on 787  degrees of freedom
AIC: 801.35

Number of Fisher Scoring iterations: 5
```

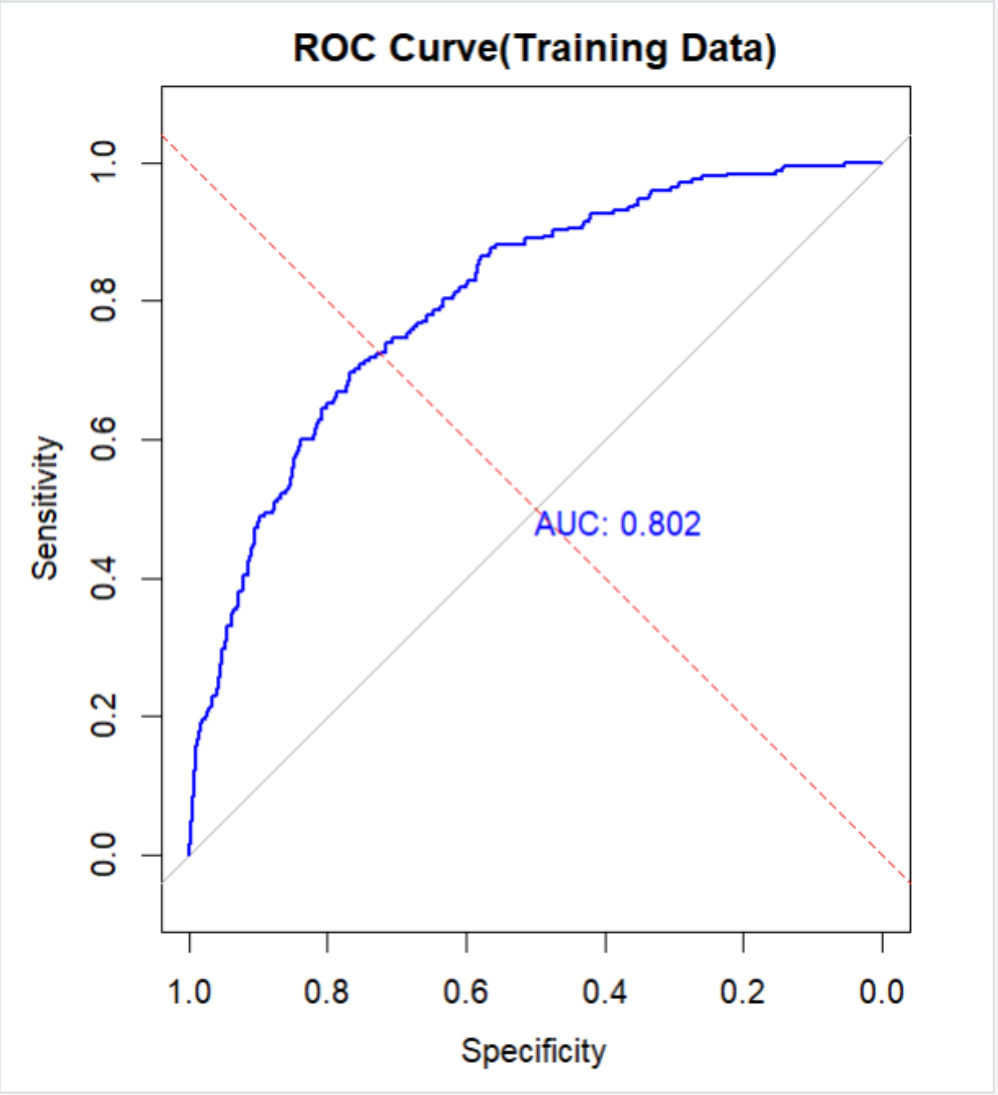
BASELINE TRAINING DATA RESULTS: 10 PRINCIPAL COMPONENTS

Prediction	Reference	
	No	Yes
No	387	62
Yes	168	183

- Accuracy: .7125
- Precision: .5214
- Recall (Sensitivity): .7469
- F1-Score: .6140
- ROC/AUC: .802

In order to fine-tune this model, we can use the threshold given by the ROC Curve to determine the cutoff for default probability

In this case, I found that threshold maximizing TPR (Sensitivity) and minimizing FPR (1 - Specificity) was .35



NEW MODEL (DEF PROB > .35 = POSITIVE)

Prediction	Reference	
	No	Yes
No	426	74
Yes	129	171

Metric	Model 1 (.30)	Model 2 (.35)	Difference
Accuracy	.713	.746	+ .033
Precision	.521	.570	+ .049
Recall	.747	.698	- .049
F1-Score	.614	.628	+ .014

While the model did improve in most metrics as expected, the trade off is not worth it.
The decrease in recall means an increase in missed defaults which is more costly than a false positive.

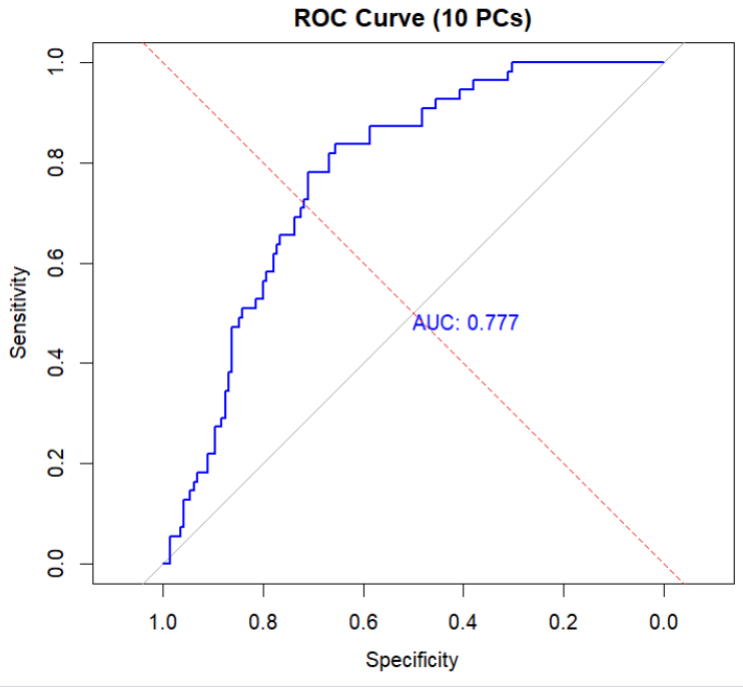
Note: In an imbalanced dataset where the number of actual positives (defaults) is very low, **recall** is a more meaningful metric than **accuracy because it measures the ability of the model to correctly identify all positive instances**

Source: (Google Developers Machine Learning Crash Course, n.d.)

MODEL RESULTS IN TEST

Prediction	Reference	
	No	Yes
No	110	19
Yes	35	36

- **Accuracy:** .725
- **Precision:** .507
- **Recall (Sensitivity):** .655
- **F1-Score:** .571
- **ROC/AUC:** .777



Using the ROC Curve. The model default threshold was adjusted to **.2484.**

This led to strong improvements in recall:

Current Recall: .7818 (+.1268)

Prediction	Reference	
	No	Yes
No	103	12
Yes	42	43

```
               Accuracy : 0.73
                 95% CI : (0.6628, 0.7902)
    No Information Rate : 0.725
    P-Value [Acc > NIR] : 0.4732

                  Kappa : 0.4209

 Mcnemar's Test P-Value : 7.933e-05

            Sensitivity : 0.7818
            Specificity : 0.7103
         Pos Pred Value : 0.5059
         Neg Pred Value : 0.8957
```

CLOSING ACT

Model Deployment and Next Steps

PROJECT OBJECTIVES

- 70% or Greater Model Accuracy: ✓
  - Actual Value: **73% overall accuracy** at baseline
  - Actual Value: **78% accurate** in identifying actual defaulters (Recall)
- Create a reproducible solution: ✓
  - **Code Repository Document** and **Process Flowchart** available for review
  - **Production Turnover Report** completed, and model turned over to IT for batch runs and maintenance
- Next Steps: ?
  - Recommendations
  - Any concerns with the current model?
  - Where can we go in the future?

RECOMMENDATIONS

- Emplyer should implement a credit application to collect as many of these details as possible prior to lending:
  - Questionnaires & Credit Reports
- Raw probability values should be utilized in credit decisions:
  - Potential Borrowers with Default Prob within +.025 of the current threshold should undergo review by the manager for approval decision (24 – 48 hours approval window)
  - They should receive a higher interest rate and a minimal credit line until creditworthiness is proven
  - All others outside the threshold will receive auto-approval
- Existing borrower detail requests should be sent out semi-yearly (6-months). New submissions should be run through the model to forecast default risk.
  - During periods of higher default risk, model thresholds can be set to more discriminant levels
  - During less risky periods, the company has the ability to lend more
- Dashboard should be created to track Performance Metrics
  - ROC/AUC > .77
  - Optimal Threshold Metrics
  - Accuracy: .70 or greater
  - Recall: .70 or greater

NEXT STEPS

- Current model exceeds parameters of initial project objectives.
  - Both in overall accuracy and recall
  - Model's threshold can be adjusted if recall drops below acceptable parameters
- Risks we are tracking at this time: Precision.
  - The model is still misclassifying just under half (precision = .51) of its default predictions.
  - This is acceptable at this time as the main goal is **preventing default**, so we'd like the model to be more discriminant.
- In future projects, model performance can be improved via other modeling methods. Current model has set a new benchmark for comparison against high consumption models.
  - Random Forests
  - AdaBoost
  - XGBoost

All of the above have been shown to exhibit improved performance over logistic regression in studies. Especially, AdaBoost in cases of default classification such as this (Lai, 2020)

“DATA IS THE SWORD OF THE 21ST CENTURY, THOSE WHO WIELD IT WELL, THE SAMURAI.”

Jonathan Rosenberg - Former Senior VP of Products, Google
Source: (Ward, 2017)

CITED SOURCES

- B. Wu, X. Yi, and D. Ning Zhao, "Generalized PCA Method and Its Application in Uncertainty Reasoning," in *IEEE Access*, vol. 13, pp. 123699-123707, 2025, doi: 10.1109/ACCESS.2019.2960261
- Google Developers Machine Learning Crash Course: ROC and AUC. (n.d.). *ROC and AUC*. Google. https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc
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